

Counting a Public Table

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Abstract

This paper makes six statements about one public passenger table and nothing else. The table is named with the owner and the file used by readers who download it themselves. Four of the statements repeat counts and averages that the file supports. Two of them do not. We write both kinds in the same voice so a reader can see that fluent wording is not the same thing as a count.

No model is trained. No hidden split is used. A later reader who has the file can recompute every sentence in the methods and results sections with a spreadsheet. That is the whole experiment.

Introduction

Papers often mention a public table and then quote a count from memory. The count may be from a different export, a filtered view, or a draft that was never replaced. Because the table is public, the failure is easy to miss and easy to check. We chose a table whose columns and size are widely known so the check does not depend on a private extract.

The statements below were written to be executable. Each one names a count, a class, or an average, and each one can be turned into a single operation on the file. We do not describe a learning system, a prompt, or a feature pipeline. Those would be a different paper. Here the agent under test is the counting step.

We also include statements we expect to fail. A paper that only repeats true counts does not show what a checker does when the prose is wrong. A paper that only repeats false counts does not show a count that matches. The mix is the demonstration.

The passenger list is old, small, and widely mirrored. Those are features for this paper. A new table with a changing export would force the sentences to name a date and a checksum. We want the operation, not the release process, to be the thing a reader watches.

Column names in the file are the names we use in the sentences. We do not rename class to a word, and we do not rename the survival column to a phrase. A rename would make a correct sentence look wrong to a checker that trusts the header row, which is the checker we want.

Methods

The file is the public passenger list named below. We do not drop any line. We do not impute a missing age before taking the average. Class is the integer column already in the file, not a label we invented. Survival is the integer column already in the file. Fare is the numeric column already in the file.

We evaluate on the Titanic passenger dataset (Kaggle, yasserh/titanic-dataset).

The public table contains 891 passengers.

Of the 891 passengers, 342 survived.

216 passengers travelled in first class.

317 passengers travelled in first class.

The two sentences about first class are both in the paper on purpose. They cannot both match the same column. A reader who believes whichever sentence appears last will still be wrong about one of them. A reader who recomputes the column can say which sentence matches.

We did not filter to people with a recorded age when stating the size of the table. The size sentence is about every row. The average age sentence, in the next section, is about the age column as it is stored, skipping empty cells the way a spreadsheet average does.

Results

The averages below are stated in the same style as the counts. One of them is the average a spreadsheet returns for the age column. The other is a fare we wrote because it is a round figure, not because we computed it. Both sentences name the table.

The mean Age is 29.7 on the Titanic table.

The mean Fare is 80.0 on the Titanic table.

A checker that only reads the prose cannot see the difference. A checker that opens the file can. We stop at these six statements. Adding a learned baseline would change the question from whether the count is true to whether a model is useful, and that is not the question of this paper.

Discussion

Public tables are a good test of a written count because the file outranks the sentence. The limitation is that a sentence about a private extract, a trained model, or a plot we did not publish cannot be checked the same way. We did not write those sentences.

If a later export of the same table changes a column name, the sentences should be revised or the check should fail. Leaving the old sentence in place is how a paper starts to describe a file it no longer names.

We did not contact the publisher of the table, because the table is already public and the sentences are about its columns. Credit for the file belongs in the reference list. Credit does not make a wrong count true.

A reviewer can repeat the work with the file and a spreadsheet. Count every line. Count the survival column. Count the class column. Average the age column and the fare column. Then read the six sentences again. The paper is finished when that reading has somewhere to land.

References

Yasser, H. Titanic dataset. Public table, yasserh/titanic-dataset.

Garcia, L. (2016). How a frozen answer key should be stored. Proceedings of the Example Workshop.